

Particles & Fields

The fundamental objects are **quantum fields**:

- lepton fields
 - quark fields
- } Fermions
- bosons fields

Particles are excitations of **quantum fields**:

- Real particles = excitation with $E^2 = \mathbf{p}^2 + \mathbf{m}^2$
- Virtual particles = excitation with $E^2 \neq \mathbf{p}^2 + \mathbf{m}^2$